WORKED SOLUTION

STEP I 2017, Question 8

Pure • Sequence

STEP I 2017, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/17-s1-q8

The key idea

The expression $a_n^2 + 2a_nb_n - b_n^2$ is unchanged by one step of the recurrence, so it stays at its starting value for ever. Dividing it by b_n^2 turns it into an exact equation for the ratio c_n, with a small error term $1/b_n^2$. Completing the square in that equation does almost all the work: it gives the limit, the side of $\sqrt{2}-1$ the ratio lies on, and the rational bounds for $\sqrt{2}$.

Solution

The induction

For $n = 1$ we have $a_1^2 + 2a_1b_1 - b_1^2 = 1 + 4 - 4 = 1$, so $(*)$ holds.

Suppose $(*)$ holds for some $n \geq 1$. Writing $a = a_n$, $b = b_n$,

$$\begin{aligned} a_{n+1}^2 &= a^2 + 4ab + 4b^2, \\ 2a_{n+1}b_{n+1} &= 2(a+2b)(2a+5b) = 4a^2 + 18ab + 20b^2, \\ b_{n+1}^2 &= 4a^2 + 20ab + 25b^2. \end{aligned}$$

Adding the first two and subtracting the third,

$$a_{n+1}^2 + 2a_{n+1}b_{n+1} - b_{n+1}^2 = a^2 + 2ab - b^2 = 1.$$

So $(*)$ holds for $n+1$, and by induction it holds for all $n \geq 1$.

Part (i)

First, every a_n and b_n is positive: this is true for $n = 1$, and the recurrence builds a_{n+1} and b_{n+1} from a_n and b_n with positive coefficients. Hence

$$b_{n+1} = 2a_n + 5b_n > 5b_n,$$

since $a_n > 0$. Starting from $b_1 = 2 = 2 \times 5^0$ and multiplying by at least 5 at each step gives $b_n \geq 2 \times 5^{n-1}$ for all $n \geq 1$ (a one-line induction).

Now divide $(*)$ by b_n^2, which is not zero:

$$c_n^2 + 2c_n - 1 = \frac{1}{b_n^2}, \qquad \text{that is} \qquad (c_n+1)^2 = 2 + \frac{1}{b_n^2}. \tag{†}$$

Ask yourself Which square root do we take?

Since $c_n > 0$, the number $c_n + 1$ is positive, so we take the positive root:

$$c_n + 1 = \sqrt{2 + \frac{1}{b_n^2}}.$$

As $n \to \infty$, $0 < 1/b_n^2 \leq \frac{1}{4}\,25^{-(n-1)} \to 0$, so $c_n + 1 \to \sqrt{2}$ (the square root function is continuous). Therefore $c_n \to \sqrt{2} - 1$, as required.

Part (ii)

From (†), $(c_n+1)^2 = 2 + 1/b_n^2 > 2$. Since $c_n + 1 > 0$ and $\sqrt{2} > 0$, this gives $c_n + 1 > \sqrt{2}$, that is, $c_n > \sqrt{2} - 1$.

For the other bound, divide 2 by both sides of $c_n + 1 > \sqrt{2}$; both sides are positive, so the inequality reverses:

$$\frac{2}{c_n+1} < \frac{2}{\sqrt{2}} = \sqrt{2}.$$

Together,

$$\frac{2}{c_n+1} < \sqrt{2} < c_n + 1.$$

Ask yourself Which value of n produces $99/70$?

We need $c_n + 1 = (a_n + b_n)/b_n$ to be $99/70$, so we look for $b_n = 70$. Running the recurrence:

$$(a_1, b_1) = (1, 2), \quad (a_2, b_2) = (5, 12), \quad (a_3, b_3) = (29, 70).$$

So $c_3 = \frac{29}{70}$ and $c_3 + 1 = \frac{99}{70}$, while

$$\frac{2}{c_3+1} = \frac{140}{99}.$$

Putting $n = 3$ into the double inequality gives

$$\frac{140}{99} < \sqrt{2} < \frac{99}{70}.$$

Checks

- $(a_3, b_3) = (29, 70)$ satisfies $(*)$: $841 + 4060 - 4900 = 1$.
- $\frac{140}{99} \approx 1.414141$ and $\frac{99}{70} \approx 1.414286$, with $\sqrt{2} \approx 1.414214$ between them. The interval has width $\frac{1}{6930}$.
- The next term, $(a_4, b_4) = (169, 408)$, gives $c_4 \approx 0.4142157$, already within 2.2×10^{-6} of $\sqrt{2} - 1$, which matches the error falling roughly like $1/b_n^2$.
- Squaring: $140^2 = 19600 < 2 \times 99^2 = 19602$ and $99^2 = 9801 > 2 \times 70^2 = 9800$.

Common slips

- Expanding $(*)$ for $n + 1$ and not collecting terms carefully: the ab and b^2 coefficients cancel only just ($4 + 18 - 20 = 2$ and $4 + 20 - 25 = -1$).
- Claiming $b_{n+1} \geq 5b_n$ without noting that $a_n > 0$.
- Forgetting to justify the positive square root, or reversing an inequality on taking reciprocals without saying both sides are positive.

Beyond STEP

The recurrence is multiplication by the matrix $\begin{pmatrix} 1 & 2 \\ 2 & 5 \end{pmatrix}$, which has determinant 1 and preserves the quadratic form $x^2 + 2xy - y^2$; finding invariant forms like this is the heart of the theory of Pell's equation. The numbers $c_n + 1$, namely $\frac{3}{2}, \frac{17}{12}, \frac{99}{70}, \ldots$, are every other continued-fraction convergent of $\sqrt{2}$; convergents are the best rational approximations for the size of their denominators. Davenport's *The Higher Arithmetic* covers continued fractions and Pell's equation beautifully.

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